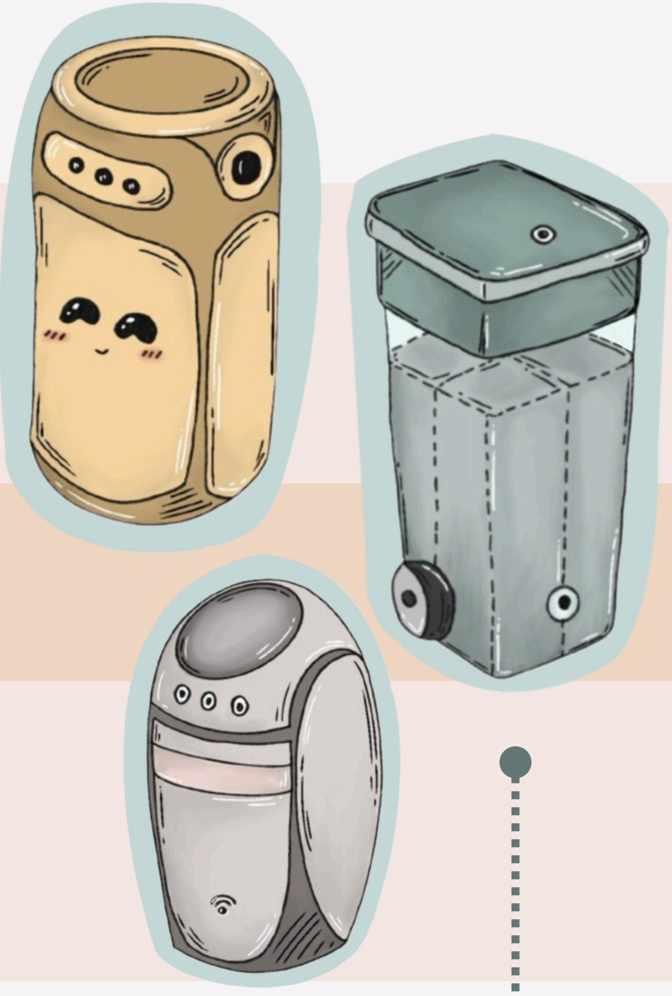
Kayla Tham Ke Yi 1008548 | Kopal Darshna 1009179 | Mehrotra Anandi 1009196 | Venkatesh Thanushree 1009339 | Audrey Dominique 1009687

Design Process

Problem Statement

How might we create an effective cleaning solution to improve hygiene standards and efficiency for cleaning staff?



1 Initial Concept Sketch

This initial sketch outlines the basic features and structure envisioned for the sorting bin.

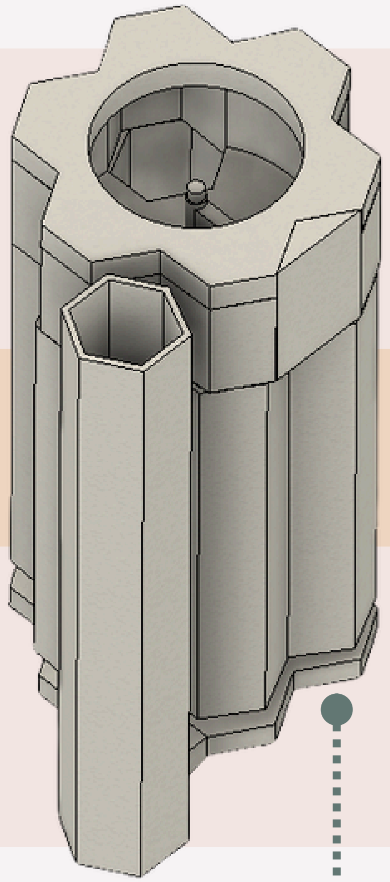
2 First Physical Iteration

Main Goal: Ensure rotation of the motors of the spinning parts are functional.

Identified Issues:

-Difficulty in rotation of the Server Motor.
-Proximity Sensor malfunctions.
-Modular compartments constantly topples.
-IR Sensor detects bin material

Each item was tested three times to ensure consistency in sorting performance. The first prototype, built with an IR sensor and basic sorting logic, showed limited success. It achieved 66% accuracy for larger plastic, 33% for smaller plastic, and 0% accuracy for both tissue and battery waste. These shortcomings were primarily due to sensor limitations, misalignment in the rotating lid, and overly simple detection logic, which led to misclassification or failure to detect certain materials.



3 First CAD ver. Model

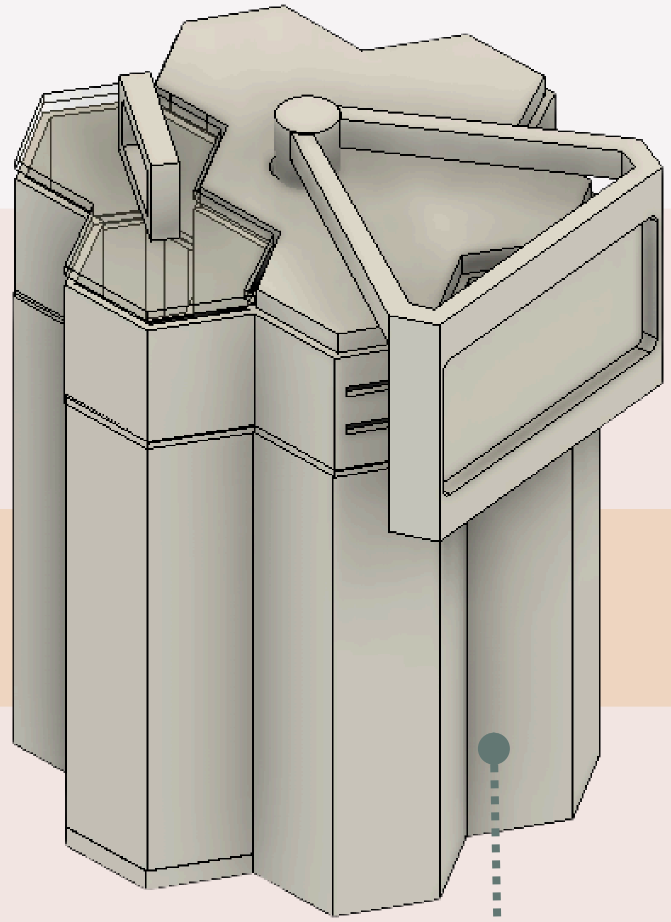
- Designed for automatic sorting after all trash is inserted
- Used hexagonal bins for a clean, modular design when detached
- Added interactive elements to enhance user engagement

4 Developed Physical Prototype

- Switched to sorting items one by one for better control and accuracy
- Replaced traditional sensors with ESP32-CAM module for better detection accuracy
- Added a transparent top cover to allow natural light into the camera module

Type of trash	Size (cm)	First Prototype Sorting Accuracy	Final Product Sorting Accuracy
Plastic	4.5 x 3	66%	**70 %**
	3 x 3	33 %	**50%**
tissue	5.5 x 4.5	0	**50%**
	4.5 x 3	0	**50%**
Battery	4.8 x 2.5	0	**50%**

Testing Accuracy Comparison: Prototype vs Final Product



5 Finalized CAD

- Extended screen display for easier interaction.
- Reusing recycled material instead of PLA

In contrast, the final prototype demonstrated 50% sorting accuracy across all tested categories, including plastic, tissue, and battery. This significant improvement was achieved by implementing several key upgrades: better mechanical alignment, replacement of the IR sensor with an ESP32-CAM module, and refined sorting algorithms. These enhancements allowed the system to reliably detect and correctly sort various waste types, making it a dependable and functional automatic trash-sorting bin.

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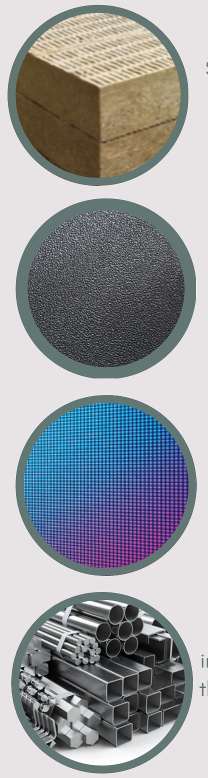
Critical Image

Material Palette

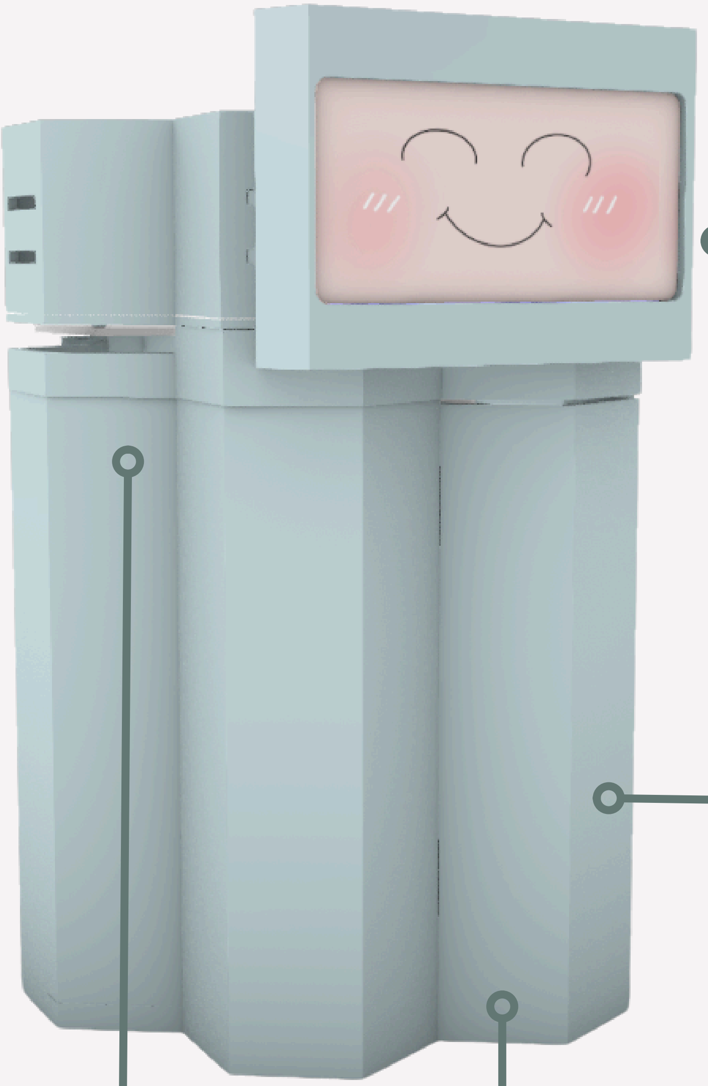


Stainless steel for nuts, bolts and gears

PLA Plastic for exterior and inner walls of bin

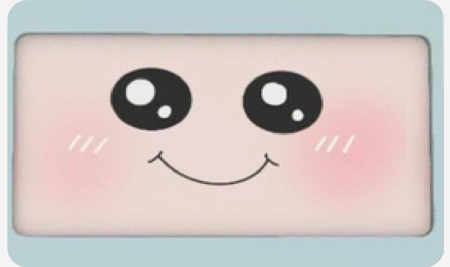
LED Panel for the Face Feature

Rock wool as insulation between the outer and inner walls of the bin



Interactive Emoticons

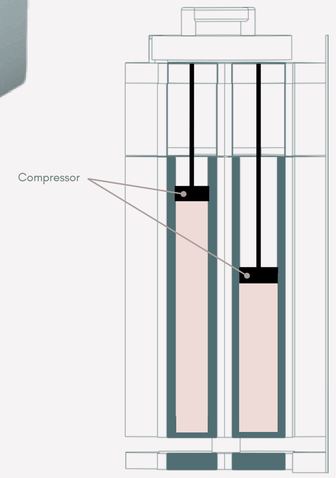
An interactive face display that reacts with facial expressions every time trash is thrown in to pique user interest.



Screen showing Default Emoticon

Detachable sides

Detachable hexagonal doors allow easier trash removal and reduce lifting effort.



Ultrasonic Sensor:

An ultrasonic sensor is mounted on the inner top surface of each bin compartment. It detects the rising trash level using sound waves and signals when the bin is full. This status is instantly updated in a connected app, alerting cleaners when a bin is full.

Compressor

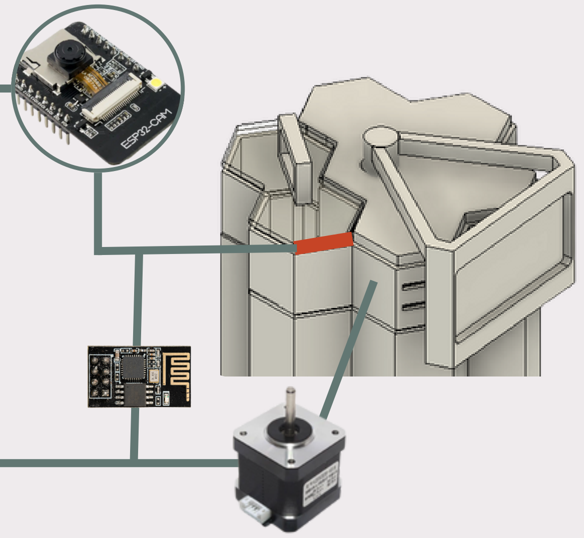
Built-in compressor in each compartment to compact trash, creating more space.

Perspective of user going to throw trash into product.

Impression of product in relation to site.

Step 1 : The sensing mechanism

Trash is thrown here. ESP32-Camera Module is placed on the side of the compartment, detecting the type of material of the trash which further sends the classified data to ESP32 microcontroller.

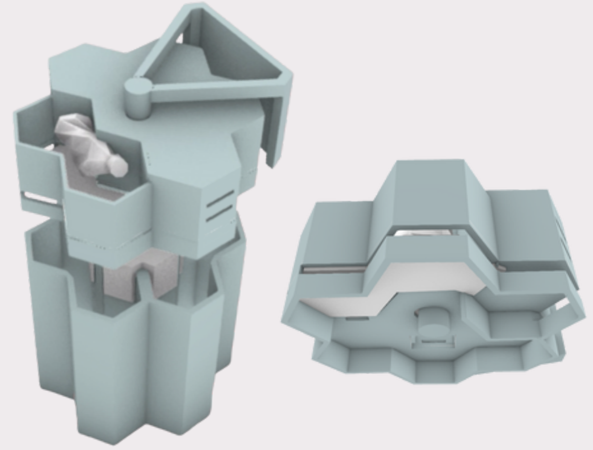


Step 2 : Trash-Based Rotation Mechanism

The second ESP32 controls a stepper motor. Based on the detected trash type, the motor rotates to align the appropriate bin compartment under the lid.

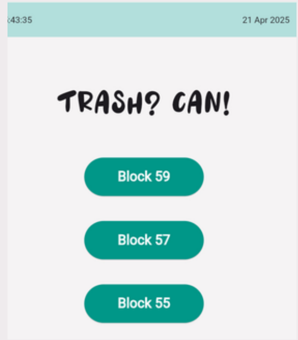
Step 3 : Trash Disposal Mechanism:

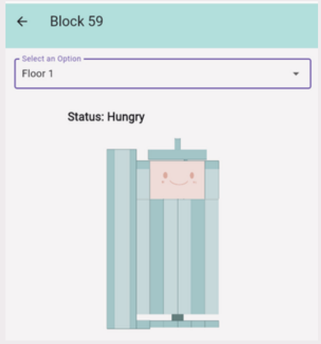
Once aligned, the trash is dropped into the correct compartment by opening the bottom lid controlled by servo motors.



Step 4 : Trash?Can! Updates

With the help of the ultrasonic sensor, Fill level of the bin is updated in the Trash?Can! app. The cleaners would receive a notification when the bin is full. Else, bin status remains as 'Hungry'






DISPENSER